# Adaptive Optimization of Public EV Charging Infrastructure in New York State

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## 1 Introduction

New York State is initiating a large-scale effort to expand its electric vehicle (EV) infrastructure as part of a broader transition toward a cleaner, more sustainable transportation system. In May 2025, state agencies formed an interagency working group to accelerate clean vehicle adoption and charging infrastructure deployment, explicitly tasked with coordinating the rapid rollout of public and private EV charging stations and leveraging approximately $1 billion in newly proposed climate funding and Regional Greenhouse Gas Initiative revenue for this purpose.[1] This working group brings together the New York State Department of Environmental Conservation, the New York State Energy Research and Development Authority, the New York Power Authority, and other state agencies to address grid reliability, streamline permitting, and make strategic use of state-owned property for charging infrastructure.

Despite these substantial commitments, planning an effective public charging network remains challenging. EV adoption is highly uneven across regions: some urban and suburban ZIP codes exhibit rapidly growing EV penetration and congestion at existing stations, while rural and disadvantaged communities face limited access and range anxiety. The state's policy goals emphasize spatial equity (serving underserved areas), system efficiency (avoiding redundant or underutilized chargers), and future readiness (accommodating uncertain growth in EV ownership and charging demand). From a planning perspective, the key decisions are:

- how to minimize total system cost;
- where existing stations should be expanded / new stations opened and how many charging ports to install;

subject to a budget requirements and constraints on charger utilization and station size.

## 2 Data Preprocessing

We draw on two complementary datasets to characterize both the existing public charging infrastructure and the spatial distribution of EV ownership in New York State. Public charging infrastructure is obtained from the U.S. Department of Energy Alternative Fuel Data Center (AFDC) Alternative Fuel Stations database, from which we retain all public EV stations located in New York State in the most recent release.[2] For each station, the data contains a unique identifier, whether it is currently in service (E = existing), temporarily unavailable (T), or planned but not yet built (P), and its geographic location (latitude and longitude). EV ownership patterns are drawn from the NYSERDA EV Registrations dataset (2011–2022), which provides vehicle-level registration records, including a vehicle identifier, registration valid date, ZIP code, and registration year.[3]

To prepare these data for the optimization model, we first standardize ZIP codes in the registration dataset to five-character, zero-padded strings and use the `pgeocode` library to retain only ZIP codes whose state code is "NY," thereby removing out-of-state registrations and malformed ZIPs. For

each registered vehicle, we sort records by registration date and retain only the most recent observation. We attach latitude and longitude centroids to each New York ZIP and treat these centroids as baseline demand locations in the model. Using the ZIP centroids and station coordinates, we compute pairwise distances via the Haversine formula to construct a distance matrix $d_{ij}$ between demand ZIP $i$ and station $j$. This distance is used to enforce geographic coverage constraints in the model.

# 3 Methodology

We use four optimization models to evaluate the charging network designs under different treatments of future demand uncertainty. All models share the same basic structure: EV charging demand is defined at the ZIP-code level, existing public stations serve as candidate sites, and decisions include whether to open or expand stations and how many charging ports to install at each site. The objective combines infrastructure costs, distance-based detour costs, and a penalty for unmet demand when drivers cannot be served within the maximum detour radius.

The general formulation of the model is as follows,

**Sets and indices**

- $i \in \mathcal{I}$: demand ZIP codes.
- $j \in \mathcal{J}$: candidate station sites (existing and potential).
- $s \in \mathcal{S}$: demand scenarios (used in models (3) and (4)).
- $\mathcal{E} \subseteq \mathcal{I} \times \mathcal{J}$: feasible ZIP–station pairs with $d_{ij} \leq R_{\max}$.

**Parameters and assumptions.** We make several assumptions about demand, capacity, costs, and detour penalties based on empirical ranges reported in recent studies of EV charging usage and infrastructure costs.[4]

- $D_i = 0.25$: expected daily charging demand at ZIP $i$, i.e., each EV requires about one charging session per day in total, of which 25% takes place at public stations.
- $D_{is}$: expected daily charging demand at ZIP $i$ in scenario $s$.
- $d_{ij}$: effective travel distance between demand ZIP $i$ and station $j$ (km).
- $R_{\max} = 15$ km: maximum allowed detour distance (used to define $\mathcal{E}$).
- $F_j = \$25{,}000$: fixed cost of opening/operating station $j$.
- $v_j = \$15{,}000$: marginal cost per charger installed at station $j$.
- $\mu_j = 3.65$: effective capacity per charger at $j$ (sessions/day), i.e., on average about one charging session every 6–7 hours per charger.
- $\bar{z}_j = 10$: upper bound on the number of chargers that can be installed at station $j$.
- $B = \$1{,}000{,}000{,}000$: total investment budget.
- $c_{\text{km}} = 0.713$: per-kilometer detour cost plus environmental cost.

- $\lambda_{\text{unmet}} = 15{,}000$: penalty per unit (session) of unmet demand.
- $M = 50$: large constant linking charger counts to station opening.
- $\pi_s$: probability of scenario $s$, with $\sum_{s\in\mathcal{S}} \pi_s = 1$.

**Decision variables**

- $y_j \in \{0,1\}$: 1 if station $j$ is open/expanded in the plan; 0 otherwise.
- $z_j \geq 0$: number of chargers installed at station $j$.
- $x_{ij} \geq 0$: fraction of demand at ZIP $i$ assigned to station $j$.
- $u_i \in [0,1]$: unmet fraction of demand at ZIP $i$.

## 3.1 2022 Baseline Model

We first solve a static optimization model using observed 2022 EV registrations. This model takes the actual 2022 ZIP–level demand $D_i^{2022}$ and optimizes which stations to open, how many chargers to install, and how to route demand subject to capacity, distance, and budget constraints. Because this model uses realized (rather than forecasted) demand, its primary purpose is to calibrate the unmet-demand penalty parameter and to verify that the core optimization framework behaves sensibly under real data. The 2022 solution is therefore not included in the comparison across 2025 deterministic, adaptive, and wait-and-see strategies, but instead provides a diagnostic baseline to ensure that the infrastructure cost structure, detour penalties, and capacity constraints are scaled appropriately. For this 2022 calibration model, we solve the following optimization problem:

$$
\begin{aligned}
\min_{x,y,z,u} \quad & \underbrace{\sum_{j\in\mathcal{J}} \big(F_j y_j + v_j z_j\big)}_{\text{infrastructure cost}} + \underbrace{\sum_{(i,j)\in\mathcal{E}} c_{\text{km}}\, d_{ij}\, D_i^{2022}\, x_{ij}}_{\text{detour/env cost}} + \underbrace{\lambda_{\text{unmet}} \sum_{i\in\mathcal{I}} D_i^{2022}\, u_i}_{\text{unmet-demand penalty}} \\
\text{s.t.} \quad & \sum_{j:(i,j)\in\mathcal{E}} x_{ij} + u_i = 1, && \forall i \in \mathcal{I}, \\
& \sum_{i:(i,j)\in\mathcal{E}} D_i^{2022}\, x_{ij} \leq \mu_j\, z_j, && \forall j \in \mathcal{J}, \\
& x_{ij} \leq y_j, && \forall (i,j) \in \mathcal{E}, \\
& z_j \leq M\, y_j, \quad z_j \leq \bar{z}_j, && \forall j \in \mathcal{J}, \\
& \sum_{j\in\mathcal{J}} \big(F_j y_j + v_j z_j\big) \leq B, \\
& x_{ij} \geq 0,\ u_i \in [0,1],\ z_j \geq 0,\ y_j \in \{0,1\}
\end{aligned}
$$

## 3.2 2025 Deterministic Optimization Model

The deterministic model uses a single point forecast $D_i^{2025}$ of daily charging demand per ZIP and optimizes both station openings and charger counts. It represents a "best-guess" expansion plan that ignores uncertainty in future EV adoption.

**Point Forecast of Demand.** To construct the point forecast $D_i^{2025}$, we first clean and reorganize the registration records into a panel of cumulative EV stock by ZIP code and year. Invalid ZIP

codes are removed, and for each ZIP we compute annual new registrations and the corresponding cumulative count of registered EVs. For every ZIP we then use only the most recent three years of data to fit a separate OLS time trend in the logarithm of cumulative stock. Extrapolating this ZIP specific trend to 2025 yields a forecast of EV stock in that year, which we convert into daily charging demand using a fixed number of charging sessions per vehicle per day.

Once $D_i^{2025}$ is obtained, we solve the deterministic optimization problem to choose $(y_j, z_j, x_{ij}, u_i)$ that minimize infrastructure cost, detour cost, and unmet demand penalties under this single point forecast. To assess robustness, the resulting first-stage decisions $(y_j, z_j)$ are then evaluated across all demand scenarios.

$$\min_{x,y,z,u} \sum_{j\in\mathcal{J}} \big(F_j y_j + v_j z_j\big) + \sum_{(i,j)\in\mathcal{E}} c_{\text{km}}\, d_{ij}\, D_i^{2025}\, x_{ij} + \lambda_{\text{unmet}} \sum_{i\in\mathcal{I}} D_i^{2025}\, u_i$$

### 3.3 2025 Adaptive Optimization Model

The adaptive model acknowledges that 2025 demand is uncertain. We construct three demand scenarios $s = 1, 2, 3$ with ZIP-level demands $D_{is}$ and probabilities $\pi_s$. First-stage decisions $(y_j, z_j)$ are chosen before the scenario is known and are therefore common across scenarios. Second-stage decisions $(x_{ijs}, u_{is})$ adapt to the realized scenario $s$. The objective minimizes expected total cost.

**Scenario-Based Forecast of Demand.** We replace the single point forecast with an explicit scenario based representation of demand uncertainty. The scenario construction has two components. First, we use the ZIP specific OLS trends described above to obtain a baseline 2025 forecast for each ZIP and then shift this forecast in log space to create low and high demand cases. The residual variation from these regressions provides a natural measure of forecast error that we use to choose the size of these shifts, so that the low and high cases correspond roughly to one standard deviation below and above the baseline forecast. This yields three demand levels $D_{is}$ at each ZIP, corresponding to low, medium, and high 2025 EV adoption.

Second, we use historical growth rates to assign probabilities to these scenarios. For each ZIP and year we compute the annual growth rate of cumulative EV stock and apply a Gaussian Mixture Model (GMM) to cluster these growth rates into three clusters. We trim extreme growth observations and tune the clustering so that the mean growth in each regime is consistent with the low, medium, and high demand levels implied by the OLS based forecasts. The share of total base year EV stock that lies in each regime is then used as the scenario probability $\pi_s$. In this way, the demand levels $D_{is}$ come from the ZIP level OLS forecasts, while the scenario weights $\pi_s$ are grounded in the empirical distribution of historical growth.

| **Scenario** | **Average Growth Rate** | **Scenario Weight** $\pi_s$ |
|---|---|---|
| Low | 0.186 | 0.306 |
| Medium | 0.336 | 0.311 |
| High | 0.507 | 0.383 |

Table 1: GMM estimates of growth-rate regimes and corresponding scenario probabilities.

With these demand scenarios and probabilities, we formulate a two-stage stochastic optimization model. The first-stage variables $(y_j, z_j)$ determine infrastructure investments, while the second-

stage variables $(x_{ijs}, u_{is})$ determine the flow of demand and unmet demand under each scenario. The complete formulation is:

$$\min_{y,z,x,u} \underbrace{\sum_{j\in\mathcal{J}} \big(F_j y_j + v_j z_j\big)}_{\text{infrastructure}} + \sum_{s\in\mathcal{S}} \pi_s \Bigg[ \underbrace{\sum_{(i,j)\in\mathcal{E}} c_{\text{km}}\, d_{ij}\, D_{is}\, x_{ijs}}_{\text{detour/environment costs in scenario } s} + \underbrace{\lambda_{\text{unmet}} \sum_{i\in\mathcal{I}} D_{is}\, u_{is}}_{\text{unmet penalty in scenario } s} \Bigg]$$

$$\begin{aligned}
\text{s.t.} \quad & \sum_{j:(i,j)\in\mathcal{E}} x_{ijs} + u_{is} = 1, && \forall i \in \mathcal{I},\ \forall s \in \mathcal{S}, \\
& \sum_{i:(i,j)\in\mathcal{E}} D_{is}\, x_{ijs} \le \mu_j\, z_j, && \forall j \in \mathcal{J},\ \forall s \in \mathcal{S}, \\
& x_{ijs} \le y_j, && \forall (i,j) \in \mathcal{E},\ \forall s \in \mathcal{S}, \\
& z_j \le M\, y_j, \quad z_j \le \bar{z}_j, && \forall j \in \mathcal{J}, \\
& \sum_{j\in\mathcal{J}} \big(F_j y_j + v_j z_j\big) \le B, \\
& x_{ijs} \ge 0,\ u_{is} \in [0,1],\ z_j \ge 0,\ y_j \in \{0,1\}.
\end{aligned}$$

### 3.4 Wait-and-see Optimization Model

The wait-and-see model assumes that the planner observes the realized 2025 demand scenario in advance and is allowed to choose a separate plan $(y_{js}, z_{js})$ for each scenario $s$. Assignments $(x_{ijs}, u_{is})$ are also scenario-specific. This yields the lowest possible expected cost and serves as an idealized benchmark for comparison with the adaptive model.

$$\min_{y,z,x,u} \sum_{s\in\mathcal{S}} \pi_s \Bigg[ \sum_{j\in\mathcal{J}} \big(F_j y_{js} + v_j z_{js}\big) + \sum_{(i,j)\in\mathcal{E}} c_{\text{km}}\, d_{ij}\, D_{is}\, x_{ijs} + \lambda_{\text{unmet}} \sum_{i\in\mathcal{I}} D_{is}\, u_{is} \Bigg]$$

$$\begin{aligned}
\text{s.t.} \quad & \sum_{j:(i,j)\in\mathcal{E}} x_{ijs} + u_{is} = 1, && \forall i \in \mathcal{I},\ \forall s \in \mathcal{S}, \\
& \sum_{i:(i,j)\in\mathcal{E}} D_{is}\, x_{ijs} \le \mu_j\, z_{js}, && \forall j \in \mathcal{J},\ \forall s \in \mathcal{S}, \\
& x_{ijs} \le y_{js}, && \forall (i,j) \in \mathcal{E},\ \forall s \in \mathcal{S}, \\
& z_{js} \le M\, y_{js}, \quad z_{js} \le \bar{z}_j, && \forall j \in \mathcal{J},\ \forall s \in \mathcal{S}, \\
& \sum_{j\in\mathcal{J}} \big(F_j y_{js} + v_j z_{js}\big) \le B, && \forall s \in \mathcal{S}, \\
& x_{ijs} \ge 0,\ u_{is} \in [0,1],\ z_{js} \ge 0,\ y_{js} \in \{0,1\}.
\end{aligned}$$

## 4 Results

We evaluate each planning model under a common experimental framework. For the 2022 baseline model, performance is computed directly under realized ZIP-level demand. For the 2025 deterministic, adaptive, and wait-and-see models, we first solve each formulation to obtain its optimal station-opening and charger-installation decisions. To ensure comparability, the deterministic

first-stage decisions are subsequently evaluated across all three 2025 demand scenarios, while the adaptive and wait-and-see models incorporate scenario information directly during optimization. For each model, we then compute expected system cost, expected served charging sessions, served fraction, the number of stations opened, and the resulting charger count, weighting scenario-level outcomes by their probabilities where appropriate. These aggregate performance metrics populate Table 2.

| Model | $R_{\max}$ (km) | Obj. cost ($bn) | Exp. served ($10^3$ sess./day) | Served fraction | Open stations | Chargers (count) |
|---|---|---|---|---|---|---|
| Baseline 2022 | 15 | 3.10 | 167.1 | 0.522 | 4660 | 45,815 |
| Deterministic 2025 | 15 | 5.10 | 184.9 | 0.398 | 5191 | 51,583 |
| Adaptive 2025 | 15 | 5.09 | 186.7 | 0.402 | 5315 | 52,918 |
| Wait-and-See 2025 | 15 | 5.06 | 186.8 | 0.402 | 5157 | 51,187 |

Table 2: Comparison of four planning models for $R_{\max} = 15$ km.

**Interpretation.** Table 2 shows that all 2025 planning models expand the network substantially relative to the 2022 baseline, increasing both the number of open stations and total charger count. However, the deterministic model delivers lower service performance, achieving the smallest served fraction (0.398) despite similar investment levels. The adaptive model improves expected service (186.7k sessions/day) with lower total cost, indicating that incorporating uncertainty leads to more effective allocation of chargers. The wait-and-see solution provides the best possible benchmark, achieving both the lowest objective value and the highest expected service, as it optimizes decisions separately for each demand realization.

To complement the aggregate results, we compute ZIP-level service fractions under the deterministic, adaptive, and wait-and-see plans. Figure 1 displays these spatial patterns, highlighting regional variation in service coverage across the three strategies.

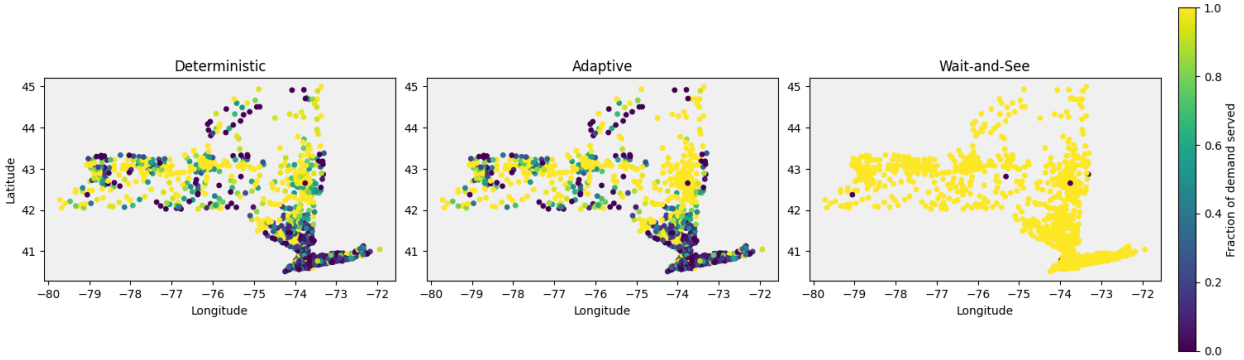


Figure 1: Spatial distribution of the fraction of EV charging demand served.

**Interpretation.** The deterministic plan exhibits several clusters of low service. The adaptive model reduces many of these gaps by reallocating infrastructure toward ZIPs with more variable demand, particularly in the north-eastern regions, leading to visibly higher coverage relative to the deterministic solution. The wait-and-see benchmark, unconstrained by non-anticipativity, achieves nearly full statewide coverage and represents the upper bound on attainable performance.

## 5 Recommendations

The four optimization models highlight important trade-offs between cost, service, and robustness. From a planning perspective, the main recommendation is to prioritize *adaptive* infrastructure expansion rather than relying on deterministic, single-forecast designs. Practically, this means using scenario-based demand projections to identify ZIP codes that remain under-served across a range of plausible futures, and allocating new stations and chargers to these robust hotspots rather than to locations favored only by the baseline forecast. The spatial maps of served fractions reinforce that adaptive planning delivers more uniform coverage across New York State and reduces the number of neighborhoods with very low access to public charging.

Table 2 shows that, relative to the 2022 baseline, the adaptive 2025 plan increases expected daily served sessions from about $167 \times 10^3$ to $187 \times 10^3$, while keeping average detours near 3 km, by expanding the network from roughly 4,660 to 5,315 stations and from 45,815 to 52,918 public charging ports. Taken together with the deterministic and wait-and-see results, we recommend that New York should target an expansion on the order of 650 additional stations and 7,000 chargers statewide by 2025, using the adaptive model to prioritize ZIP codes where this capacity will most effectively raise the fraction of demand served.

Finally, even the wait-and-see benchmark leaves four ZIP codes unserved in Figure 1. These cases are not performance failures of the optimization models but structural limitations of the existing candidate-site network. Three of the ZIP codes lie in sparsely populated rural areas of upstate and western New York, where no candidate site falls within the allowed 15 km detour radius. The remaining ZIP corresponds to a state government complex rather than a residential community, so it does not represent a genuine charging demand center. Because the model can only assign demand to locations within this coverage radius, these ZIPs remain unserviceable under all strategies. This pattern suggests a planning recommendation: future expansions should introduce new candidate sites in rural regions, where geographic isolation, rather than insufficient capacity, prevents full coverage of statewide charging demand.

## 6 Conclusion

This project builds and compares a sequence of optimization models for planning New York State's public EV charging network. Starting from a baseline model, it formulates deterministic, adaptive two-stage stochastic, and wait-and-see models that share a common network design structure but differ in how they treat demand uncertainty and information timing. These formulations are implemented and solved on ZIP-level EV registration and station data, yielding quantitative benchmarks for system cost, service coverage, and robustness. Together, the models provide a coherent optimization-based framework for evaluating alternative infrastructure expansion strategies under realistic data and policy constraints.

Looking ahead, several extensions could help strengthen the analysis. First, the current model treats demand and charger utilization in a highly aggregate way, so future work could incorporate richer behavioral responses such as price-sensitive charging, time-of-day preferences, and substitution between home, workplace, and public charging to better capture how drivers actually use the network. Second, incorporating local grid constraints, for example limits on how much additional load nearby distribution lines and transformers can safely support, would make it possible to

identify locations where otherwise attractive charging expansions are infeasible or would require substantial grid upgrades. Third, a dynamic, multi-period planning model could be developed to jointly optimize the timing of investments over a longer horizon and to evaluate robustness to deeper uncertainty in electric vehicle adoption, technology costs, and policy incentives.

# A References


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